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A spiking neural network inspired by neuroscience and psychology for Western mode- and key-conditioned music learning and composition

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Musical mode is a fundamental element of tonal music, structuring pitch organization and shaping tonal relationships. Existing artificial intelligence approaches to symbolic music generation often rely on rigid alignment strategies and simplified tonal representations, limiting their ability to capture the diversity of musical modes, in contrast to the complex perceptual and learning mechanisms observed in human listeners. In this paper, we propose a brain-inspired spiking neural network that integrates biologically grounded mechanisms with symbolic music theory to represent and learn musical modes and keys. The model comprises multiple interacting subsystems inspired by the functional organization of relevant brain regions, and incorporates neural circuit evolution and spike-timing-dependent plasticity to support mode- and key-conditioned music learning and generation. Experimental results show that the synaptic connectivity patterns emerging in the proposed network exhibit strong alignment with the Krumhansl-Schmuckler key profiles, a well-established model of tonal perception in music psychology. Additionally, quantitative evaluations show that the generated musical pieces preserve tonal characteristics while maintaining melodic diversity. By integrating insights from neuroscience, music psychology, and music theory within a spiking neural network framework, this work provides an interpretable and biologically inspired approach to symbolic music learning and generation.

Keywords Brain-inspired spiking neural network, Mode perception, Symbolic music learning, Music memory, Music generation

Music is intrinsic to human nature, engaging emotions, cognition, and cultural identity, and has long attracted researchers across disciplines, including neuroscience, psychology, and artificial intelligence (AI). In recent years, AI models have demonstrated impressive performance in symbolic music learning and generation through deep learning architectures, such as recurrent neural networks (RNN)^{1–8}, variational autoencoders (VAE)^{9–11}, generative adversarial networks (GAN)^{12–14}, and Transformers^{7,8,15,16}, even large language models (LLMs)^{17–19}. Despite these advances, existing models often treat tonal frameworks such as modes and keys through rigid alignment strategies, for example by transposing all pieces into a canonical key (C major or C minor)^{6,13}, and through simplified tonal representations⁸, typically encoding key information as flat categorical labels. Such treatments limit their ability to capture the diversity of musical modes. In addition, their black-box nature is widely argued to lack biological plausibility and offer limited interpretability^{20–22}, which further constrains their capacity to form structured, interpretable representations of musical knowledge.

As fundamental elements of tonal music, tonal frameworks such as modes and keys provide structured organizations of pitch relations and support stable tonal perception across musical contexts. This organiza-

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tion is captured by the **Krumhansl–Schmuckler (KS)** psychological model^{23,24}, which characterizes tonal hierarchies within a given key based on listeners' ratings of pitch stability, and has become a foundational reference for tonal perception research in psychology and cognitive science. Such tonal frameworks have been widely described as internalized and abstract organizational structures²³. From a cognitive perspective, this type of internalized and structured knowledge can be understood as schematic knowledge, namely abstract representations that organize prior knowledge and guide perception by constraining how incoming musical events are interpreted²⁵, while also supporting memory by structuring how sequential information is encoded and later retrieved²⁶. At the neural level, schematic processing is strongly associated with the medial prefrontal cortex (mPFC) and its dynamic interaction with hippocampal memory systems²⁷, supporting rapid learning, robust memory formation, and the retrieval of structured knowledge^{28,29}. In addition, music cognition relies on memory systems for maintaining and retrieving musical information over time, with the hippocampus, medial temporal lobe (MTL), and prefrontal cortex (PFC) implicated in temporal context memory and sequence processing^{30–33}. In parallel, music perception depends critically on the auditory system, with the auditory cortex supporting pitch and frequency encoding^{34–37} and additional neural populations contributing to temporal and rhythmic processing^{38–40}.

Box 1 | Abbreviation List

DL - Deep learning, a class of machine learning methods that employ neural network architectures to automatically learn hierarchical feature representations from large amounts of data, enabling complex pattern recognition and generative modeling.

ANN – Artificial Neural Network, a computational learning model composed of layers of interconnected nodes (neurons) that adjust weighted connections to approximate complex functions, supporting tasks such as classification, prediction, and pattern recognition.

CNN - Convolutional Neural Network, a type of ANN models that captures local patterns in data through convolutional filters.

RNN – Recurrent Neural Network, a class of ANN architectures designed to capture sequential dependencies in data.

LSTM - Long Short-Term Memory, Long Short-Term Memory, a classic variant of recurrent neural networks (RNNs), widely used for modeling long-range dependencies in sequential data.

VAE – Variational Autoencoder, a probabilistic generative model that learns latent representations for data generation.

GAN – Generative Adversarial Network, a generative framework consisting of a generator and a discriminator in competition.

SNN – Spiking Neural Network, brain-inspired computational model in which neurons communicate via discrete spike events over time, rather than continuous activations. By incorporating temporal dynamics and biologically plausible learning rules.

KS model – Krumhansl–Schmuckler model, a psychological model for tonal perception and key estimation.

TS – Tonal Subsystem, a component in our model responsible for encoding mode and key knowledge.

SMS – Sequential Memory Subsystem, a component that encodes and memorizes ordered pitch and duration information.

STDP - Spike-timing-dependent plasticity, a biologically-plausible synaptic learning rule where the strength of a synapse is adjusted based on the precise timing between pre- and postsynaptic spikes: presynaptic spikes arriving shortly before postsynaptic firing strengthen the connection, whereas the reverse order weakens it.

SHTE – The proposed symbolic music dataset in this paper comprising 193 four-part harmony excerpts from Sposobin's Harmony Textbook Exercises.

Bach – A baseline symbolic dataset consisting of 408 famous J.S. Bach's four-part chorales.

Despite progress in understanding the neural and psychological basis of tonal perception, most mainstream deep learning approaches to symbolic music generation do not explicitly incorporate biologically motivated mechanisms or temporally grounded learning rules that could support such structured representations. In this context, spiking neural networks (SNNs) offer a closer match to brain computation. Specifically, SNNs employ biologically realistic neuron models with temporal spiking behavior, and their spike-based coding and spike-timing-dependent plasticity (STDP) naturally support learning from the timing and order of musical events, which is essential for sequential music processing. Moreover, their adaptive synaptic plasticity allows tonal schemas to be flexibly updated with new experiences, providing a computational basis for learning structured tonal regularities. Taken together, these properties make SNNs a suitable candidate for linking biologically motivated learning mechanisms with psychological-level tonal organization in symbolic music generation. In our previous studies, we proposed spiking neural network models inspired by cortical structures for music memorization⁴¹,

as well as stylistic⁴² and emotional⁴³ melody generation. However, these works did not incorporate explicit tonal-theoretical structures or evaluate learned representations against established psychological models.

In this paper, we take Western tonal theory as a foundational research point and propose a brain-inspired spiking neural network model that learns and generates multi-track music by integrating brain mechanisms with psychological models of music cognition. The main contributions are as follows:

- We developed a multi-area collaborative model¹ based on a Spiking Neural Network (SNN) within the open platform **BrainCog**⁴³ to perceive, learn, and generate four-part symbolic music. The model incorporates a hierarchical tonal subsystem and a sequential memory subsystem, inspired respectively by the medial prefrontal cortex (mPFC) and auditory cortex, to encode and learn musical features, including Western modes, keys, pitch, and duration, along with their interrelationships. To ensure biological plausibility and adaptability, we adopt the Izhikevich neuron model and apply synaptic creation together with spike-timing-dependent plasticity (STDP), enabling dynamic circuit evolution.
- Experimental results demonstrate that our approach, guided by biologically inspired plasticity mechanisms, enables the model to evolve internal representations that exhibit a strong similarity to human tonal perception patterns, specifically aligning with the tonal hierarchies formalized by the **Krumhansl-Schmuckler** model. For clarity, the musical terms used in this paper are provided in Supplementary Table S1 online.
- We introduce a novel symbolic dataset, SHTE, which is suitable for tonal structure research, and establish a comprehensive framework for the quantitative evaluation of music generation. The results reveal that the generated samples effectively capture the tonal characteristics of different modes and keys, further supporting the efficacy of our approach in generating music with tonal structures that align with the **Krumhansl-Schmuckler** psychological model.

Methods

Model architecture

In this study, we utilize symbolic representations of musical pieces as the dataset and introduce a brain-inspired model based on a spiking neural network, drawing inspiration from recent advancements in neuroscience and psychology within the music domain. As illustrated in Fig. 1, the music information is prepared in a symbolic manner and divided into two parts: (1) treating the mode and key of a musical piece as theoretical musical knowledge, and (2) describing ordered notes, along with their respective pitches and durations with MIDI standard (Musical Instrument Digital Interface, a widely adopted protocol that encodes musical events as numerical values for digital processing). Then, the model centers around two key components: a tonal subsystem that embeds modes and keys as foundational prior knowledge directing the subsequent learning task, and a sequential memory subsystem designed to learn and store the ordered notes of music pieces.

¹Implementation available at https://github.com/BrainCog-X/Brain-Cog/tree/main/examples/Knowledge_Representation_and_Reasoning/musicMemory/task

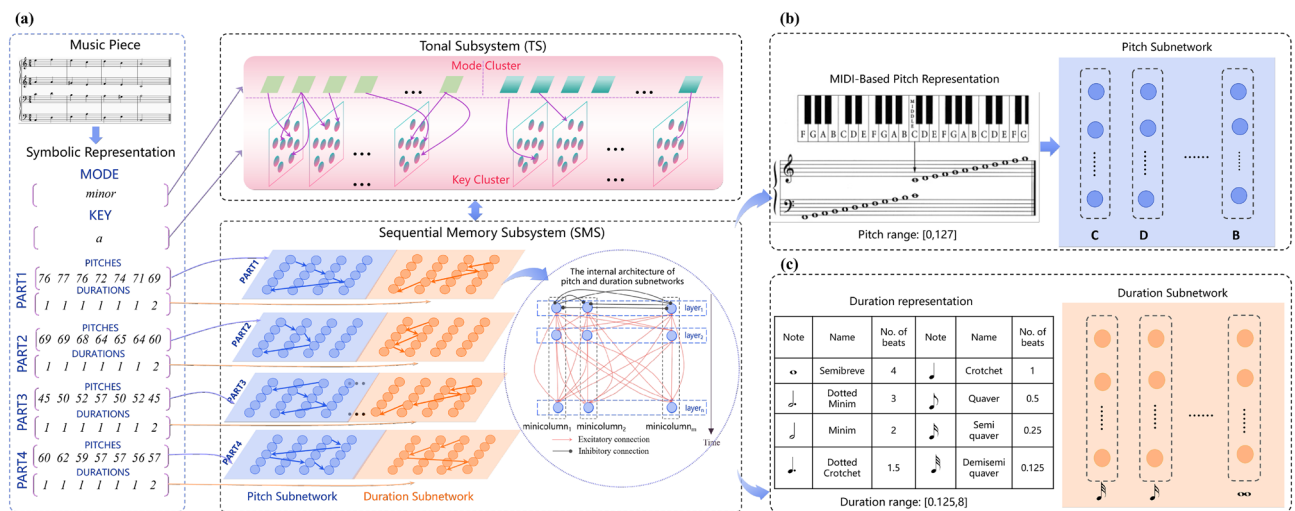


Fig. 1. The overall architecture of the proposed model. **(a)** The model integrates the tonal subsystem (TS) and the sequential memory subsystem (SMS). The tonal subsystem includes the mode cluster and key clusters, which are responsible for encoding the related music knowledge. The SMS receives the symbolic representation of the pitches and the durations, encoding and memorizing the relationships of the ordered notes. **(b)** The pitch subnetwork comprises 128 minicolumns covering the full MIDI range (0–127). **(c)** The duration subnetwork comprises 64 minicolumns, each representing a discrete time bin from a demisemiquaver (0.125) to two semibreves (8), capturing typical time intervals in common musical contexts.

Tonal subsystem

As discussed in the Introduction Section, tonal frameworks such as modes and keys have been widely characterized as internalized, abstract organizational structures in music cognition, and can be viewed as forms of schematic knowledge from a functional perspective²³. Schema representations are commonly associated with medial prefrontal cortex (mPFC) function and its interaction with the hippocampus, supporting the integration, learning, and retrieval of structured prior knowledge across experiences^{25,27–29}. Inspired by the functional role of the mPFC, the tonal subsystem is designed as a hierarchical structure that encodes modes and keys to guide subsequent learning and generation processes. As shown in Fig. 2C, the first layer, the mode cluster, contains two neural groups dedicated to encoding the Western major and the minor modes. Each group in this layer consists of twelve neurons corresponding to the tones in 12-Tone Equal Temperament (12-TET), with seven of them representing the tones (I to VII) within the pattern and the rest ones encoding those outside of the pattern, Fig. 2A illustrates the details of the major mode representation. The second layer, called the key cluster, consists of 24 neural groups, each encoding 12 major and 12 minor keys. Similarly, each group comprises twelve notes with a specific tonic, Fig. 2B shows how the neural group encodes the G major. Synaptic connections (shown in Fig. 2C) are projected from neurons in each group in the first layer that represent the tone scale degree of the mode to those in the second layer encoding the corresponding note scale degree.

Sequential memory subsystem

A musical piece is composed of an ordered sequence of notes distributed across multiple instrumental parts or tracks, each playing a distinct harmonic role and collectively contributing to the overarching musical texture. To preserve the unique characteristics of each voice, the sequential memory subsystem (SMS) is organized into four parts, corresponding to the four typical voices in polyphonic music: Soprano, Alto, Tenor, and Bass. This architectural design was originally proposed in our previous work⁴² and is adopted here to ensure structural consistency. As illustrated in Fig. 1, each part comprises a dual-network configuration: a pitch subnetwork that encodes the tonal information of notes, and a duration subnetwork that encodes the temporal intervals specifying how long each note is sustained.

The pitch subnetwork Inspired by the neural populations in the primary auditory cortex that respond distinctively to different frequencies^{34–37}, the pitch subnetwork is proposed to consist of 128 functional minicolumns as its building blocks. As depicted in Fig. 1, each minicolumn is configured as a vertical column, representing one of the 128 pitches in accordance with the MIDI standard. Each minicolumn comprises numerous neurons, all of which share a common preference for a specific MIDI pitch index. For instance,

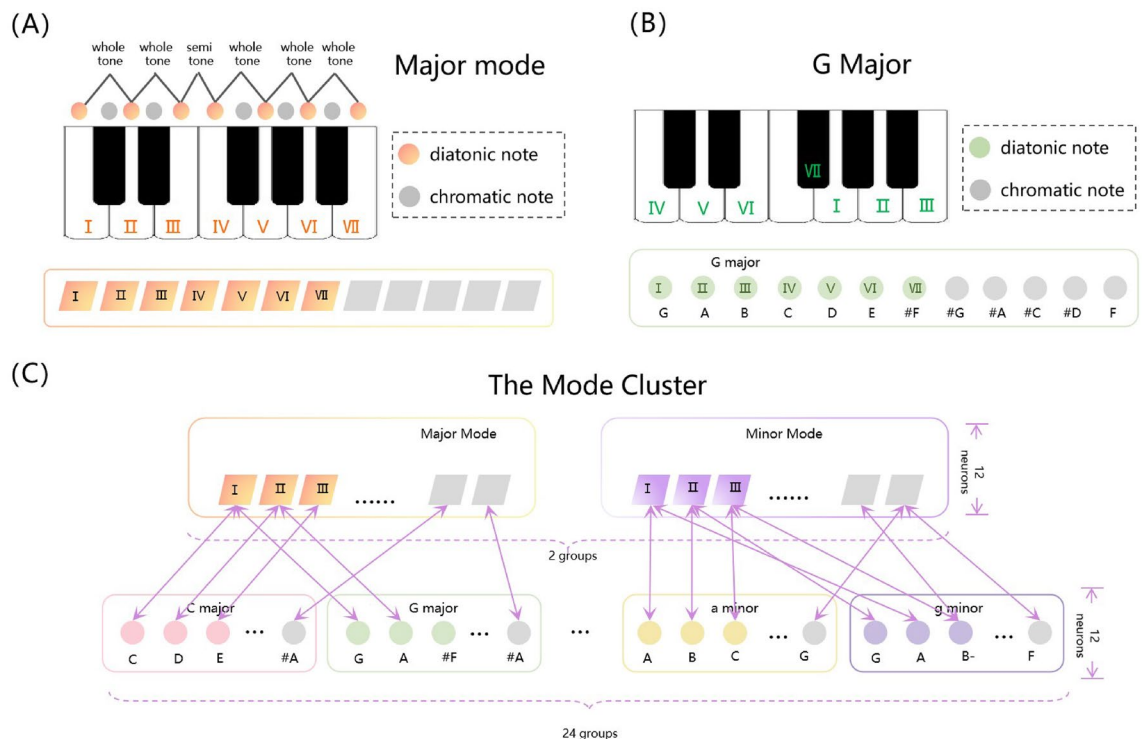


Fig. 2. The tonal subsystem, (A) describes how a neuron group in the first layer in the mode cluster represents the major mode. Seven neurons drawn by orange parallelograms encodes the diatonic tones and the gray ones encodes the chromatic tones; (B) employs the key of G major to illustrate the principle, neurons drawn by green parallelograms encodes diatonic tones, G, A, B, C, D, E, #F, the gray parallelograms also encodes the rest tones; (C) draws the hierarchical connection architecture of the mode cluster.

all neurons within the minicolumn indexed at 60 will respond preferentially to the pitch C4. The synaptic connections within and between the minicolumns are detailed below.

The duration subnetwork Inspired by neuroscientific studies showing that certain populations of cortical neurons respond selectively to specific temporal intervals in the millisecond range³⁸, the duration subnetwork is designed to represent note durations in a biologically plausible and musically meaningful way, which was originally developed in our previous work⁴¹. Its internal architecture is identical to that of the pitch subnetwork, but each minicolumn encodes a different temporal interval. The duration subnetwork consists of 64 functional minicolumns that span a range of note durations from a demisemiquaver (1/32 note) to two semibreves, which empirically covers the majority of note lengths encountered in common musical contexts. Each minicolumn corresponds to a discrete duration bin, with durations encoded on a fixed relative scale—for example, a crotchet is assigned a value of 1.0, a quaver 0.5 and a semibreve 4.0. This representation enables the model to capture rhythmic structures as a sequence of categorical temporal intervals. While duration is inherently continuous, this discretized encoding strikes a balance between biological plausibility and computational tractability.

Intra-connection As illustrated in Fig. 1, the pitch and duration subnetworks share the same internal connection structure. Inspired by the excitatory and inhibitory synapses in the brain, the connections between adjacent layers and across layers in our model are excitatory and fully connected, primarily serving to strengthen links between neurons and facilitate the flow of activity. Synaptic plasticity and transmission delay are also incorporated as crucial mechanisms in the learning process. Synaptic plasticity enables the adjustment of connection weights based on experience, mirroring the adaptive capabilities of biological neural networks, while transmission delay accounts for the time required for signals to travel between neurons, ensuring accurate representation of neural firing dynamics. By contrast, connections within the same layer are inhibitory (lateral inhibition), emphasizing neuronal competition and contributing to stable network dynamics. For further details about the sequential memory system, see previous works^{41,42}.

Music data encoding

The encoding process aims to transform external stimuli into neural spikes, which are discrete electrical events triggered when a neuron's membrane potential surpasses a firing threshold, and serve as fundamental units for encoding and transmitting information in neural systems. Suppose a music piece consisting of multiple parts is defined as $NS = \{N_{i,j} | i = 1, 2, \dots, n_j, j = 1, 2, 3, 4\}$, where $N_{i,j}$ denotes the note at i -th position in the j -th part, n_j refers to the number of notes in the j -th part. The mode can be written as M_r , where $r = 0$ or $r = 1$ refers to the major or minor mode. The key is marked as K_s , where $s \in [1, 12]$ denotes one of the 12 possible tonal centers. Then, the encoding process can be divided into two steps:

Step1: Transformation of external stimuli The external stimuli (such as keys, pitches, etc.) are transformed into the input current I that is suitable for the computational neuron model to receive. For example, a G major musical piece activates the neurons in the major cluster in the first layer and G major in the second layer. Equations (1)–(2) describes the transformation for the mode and key clusters, respectively.

$$I_{E_{-i}}^{M_r}(t) = \alpha^{M_r} \delta(x_{ij}(t) - sd_i^{M_r}) \quad (1)$$

$$I_{E_{-i}}^{K_s}(t) = \alpha^{K_s} \delta(x_{ij}(t) - ts_i^{K_s}) \quad (2)$$

where $I_{E_{-i}}^{M_r}(t)$ is the input current for neuron i in the r -th group of the mode cluster at time step t . $x_{ij}(t)$ represents the external stimuli, which refers to the pitch of the input note $N_{i,j}$ at time t . $sd_i^{M_r}$ denotes the scale degree of the tone which the neuron i represents in the mode cluster. The Dirac delta function $\delta(\cdot)$ serves as a binary indicator. Similarly, the $I_{E_{-i}}^{K_s}(t)$ refers to the input current of neuron i in the s -th group of the key cluster, $ts_i^{K_s}$ denotes the tone scale in G major cluster which neuron i represents. $\alpha^{M_r} = \alpha^{K_s} = 50$ are the scale factors to control the input values.

The sequential memory subsystem is responsible for transforming the pitch and duration of ordered notes by the pitch and duration subnetworks in the j -th part, the current are as follows:

$$I_{E_{-ij}}^P(t) = \alpha^P \delta(x_{ij}(t) - p_{hj}) \quad (3)$$

$$I_{E_{-ij}}^D(t) = \alpha^D \delta(y_{ij}(t) - d_{kj}) \quad (4)$$

where $x_{ij}(t)$ and $y_{ij}(t)$ are the pitch and duration of the note $N_{i,j}$ at time step t , p_{hj} and d_{kj} denotes the preferences of the neurons in the h -th and k -th minicolumns in the pitch and duration subnetworks of part j , respectively. $\alpha^P = \alpha^D = 30$ controls the scale of the current. Step2: Neural spiking In this study, we use the Izhikevich neural model⁴⁴ to simulate the behaviors of neurons within both the tonal subsystem and the sequential memory system. The model is described by the following equations:

$$\begin{aligned} v(t+1) &= \begin{cases} 0.04v(t)^2 + 5v(t) + 140 - u(t) + I(t), & v(t) < V_{th} \\ c, & v(t) \geq V_{th} \end{cases} \\ u(t+1) &= \begin{cases} a(bv(t) - u(t)), & v(t) < V_{th} \\ u(t) + d, & v(t) \geq V_{th} \end{cases} \\ S(t) &= \begin{cases} 0, & v(t) < V_{th} \\ 1, & v(t) \geq V_{th} \end{cases} \end{aligned} \quad (5)$$

where, $v(t+1)$ and $v(t)$ represent the membrane potential at time $t+1$ and t , $u(t+1)$ and $u(t)$ denote the recovery variable at time $t+1$ and t , respectively. $I(t)$ is the synaptic current input to the neuron at t . a , b , c , and d are parameters that control the model to fire with different spiking patterns. When the membrane potential reaches the threshold V_{th} , it is reset to c and the recovery variable $u(t)$ is incremented by d . $S(t)$ represents whether the neuron exhibits a spike at time t . Neurons in the tonal and sequential memory subsystems are simulated by the Izhikevich neural model in response to the input current I and deliver the spikes. In this study, we set the parameters $a = 0.1$, $b = 0.2$, $c = -65$, $d = 30$, $V_{th} = 30$.

Learning based on neural circuits evolution

The brain utilizes multiple regions to work together, with dynamic synaptic formation and elimination creating diverse neural circuits to accomplish various cognitive tasks. Inspired by these mechanisms, we consider the learning process as a collaborative effort among interconnected subnetworks, with a focus on the dynamic neural circuit evolution over time.

Synaptic creation

Since the music theory system stores prior knowledge, the interconnected architecture is preset. However, there are no synaptic projections between the tonal subsystem and the sequential memory system at the initial state.

Neuroscientific research has demonstrated that neural electrical activities and the dynamics of axonal growth cone movements are interdependent and coordinated. Electrical activities serve as feedback signals for the navigation of growth cones, while the movement of growth cones and the formation of new synapses subsequently influence the electrical activities within the neural network^{45–47}. In this study, we simplify this complex process by establishing rules for the formation of new synaptic connections and neural circuits in response to the input of musical information. The rule of synaptic creation is described as Eqs. (6) and (7).

$$o = \sum_f \sum_n \delta(t_i^f - t_j^n) \quad (6)$$

$$c_{ij} = \begin{cases} 1, & o \geq 5 \\ 0, & \text{else} \end{cases} \quad (7)$$

where, the t_i^f and t_j^n represent the spike time f and n of postsynaptic neuron i and presynaptic neuron j , respectively. When the oscillatory times $o \geq 5$ of these neurons i and j , a new synaptic connection is formed, and is represented as c_{ij} . This simple rule is defined to describe that neurons that oscillate together are more likely to form a synaptic connection between them.

The establishment of novel neural pathways mainly occurs between the tonal subsystem and the sequential memory subsystem, which indicates the collaborative learning and interaction of these two subsystems. Figure 3 illustrates a simple example of how the model learns a four-part music piece in A minor. The model starts by receiving a set of notes $N_1 = \{N_{1,1}, N_{1,2}, N_{1,3}, N_{1,4}\}$ from four parts, accompanied by their respective

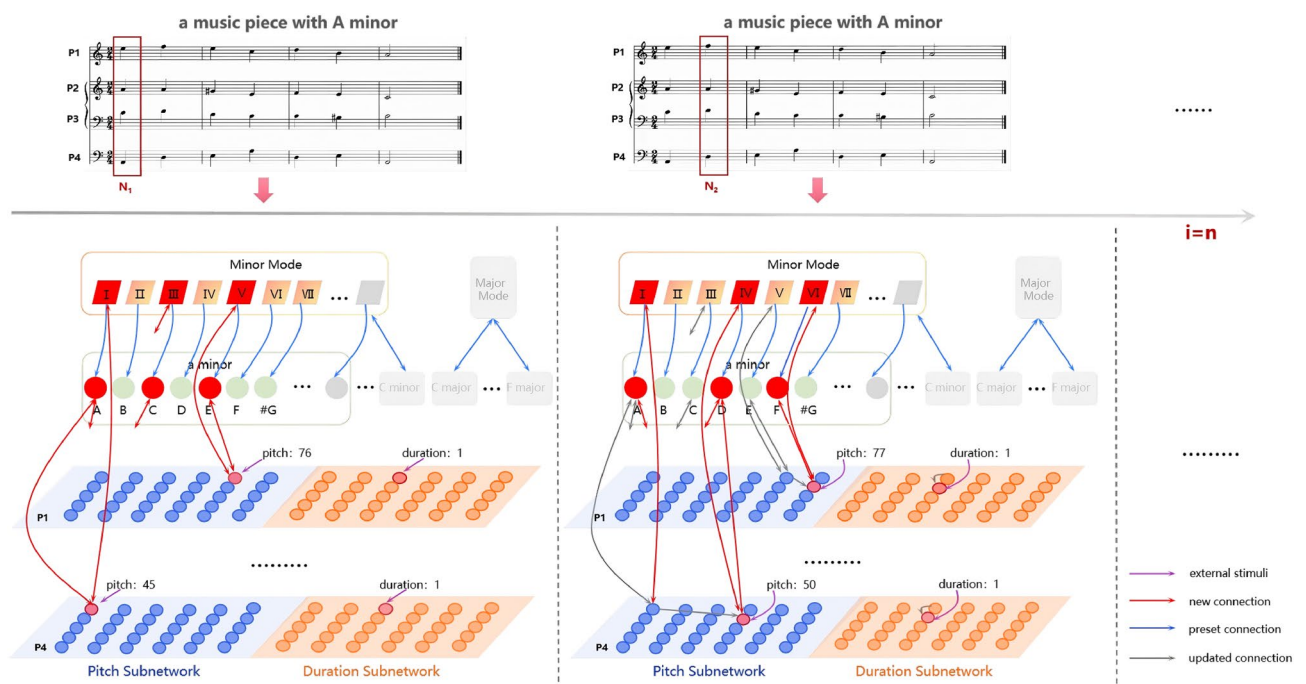


Fig. 3. The learning process of music guided by the mode theory.

symbolic pitches $X_1 = \{76, 69, 60, 45\}$. These notes fire the corresponding neurons in the mode and key clusters (highlighted by red parallelograms and circles) according to Eqs. (1)–(2) and Eq. (5). Simultaneously, the notes N_1 also prompt the activation of neurons (represented by red circles) situated across various minicolumns within the pitch subnetworks of the sequential memory subsystem. These neural oscillations facilitate the formation of both feedforward and feedback connections (marked as red double arrows) between neurons in the tonal subsystem and those in the sequential memory system by the rule (6). The synaptic weights are initially set at random values, reflecting the stochastic nature of early learning stages. These weights will be subsequently updated by the STDP learning rule as the learning process unfolds. The transmission delay del for these new connections is set to zero in this phase, implying instantaneous signaling for simplicity. It is worth noting that the prior knowledge relationships, embodied as synaptic connections, are already established and depicted by blue arrows. Upon receiving the subsequent note set N_2 , a similar pattern ensues, triggering the activation of corresponding neurons across both subsystems. When no connections link two oscillating neurons, novel connections are established spontaneously. This intricate process is characterized by the creation of neural circuits spanning the two subsystems.

Synaptic plasticity

This paper utilizes the modified STDP (Spike-Timing Dependence Plasticity)⁴⁸ learning rule to account for the spiking transmission delay when updating synaptic weights during the learning process. STDP is a biologically inspired mechanism in which the strength of a synapse is adjusted based on the relative timing of spikes from pre- and post-synaptic neurons, enabling the network to capture temporal correlations in neural activity, as shown in Eq. (8).

$$\Delta w_j = \sum_{f=1}^N \sum_{n=1}^N W(t_i^f - t_j^n - t_{i,j}^{del})$$

$$W(\Delta t) = \begin{cases} A^+ e^{-\frac{\Delta t}{\tau_+}} & \text{if } \Delta t > 0 \\ -A^- e^{\frac{\Delta t}{\tau_-}} & \text{if } \Delta t < 0 \end{cases} \quad (8)$$

where Δw_j is modulate weight of the synapse j . t_i^f and t_j^n denotes the spike time of post-synaptic neuron i and pre-synaptic neuron j , respectively. $t_{i,j}^{del}$ represents the axonal transmission delay between this neuron pair. A_+ , A_- are scale factors, and τ_+ , τ_- are time constants.

As depicted in Fig. 3, following the emission of spikes by neurons within the pitch and duration subnetworks triggered by the input of the initial note set N_1 , which comprises a pitch set $X_1 = \{76, 69, 60, 45\}$ and a duration set $Y_1 = \{1, 1, 1, 1\}$, the model proceeds to process the subsequent note set N_2 . Upon arrival of N_2 , characterized by its pitch set $X_2 = \{77, 69, 62, 50\}$ and identical duration set $Y_2 = \{1, 1, 1, 1\}$, the corresponding neurons generate spikes in distinct minicolumns, each with its own preference for different parts of the input (highlighted by red circles). It becomes evident that neurons in the sequential memory system receive inputs not just from the external stimuli, but also from other layers within the same subnetworks and from the tonal subsystem, integrating this information for further processing. Then, the inputs of the i -th neuron in the j -th minicolumn in the pitch subnetwork can be described as Eq. (9).

$$I_{ij}^P(t+1) = I_{E-ij}^P(t) + \sum_s w_{si}^{TS}(t) + \sum_r w_{ri}^P(t) \quad (9)$$

$I_{ij}^P(t)$ denotes the total input of the i -th neuron in j -th minicolumn located in the pitch subnetwork at time step t . $I_{E-ij}^P(t)$ is the current caused by the external stimuli (pitch value), w_{si}^{TS} means the s -th synaptic weight from the tonal subsystem, and w_{ri}^P represents the r -th synaptic weight from the neurons in pitch subnetwork. The detailed discussion regarding this component can be found in our earlier work⁴¹. Similarly, the input of neurons in the duration subnetwork can be expressed by Eq. (10). A notable difference, however, is that neurons in the duration subnetwork do not receive the signals from the tonal subsystems.

$$I_{ij}^D(t+1) = I_{E-ij}^D(t) + \sum_r w_{ri}^D(t) \quad (10)$$

Mode-conditioned music generation

Specifying the mode and key is a fundamental step in the process of composing a piece of music. The model described in this paper requires not only the mode and key but also a set of seed notes to initiate the creative process. As illustrated in Fig. 4, the model is tasked with generating a four-part musical piece in G minor, starting with the tonic chord (G2-Bb3-D4-G4) as the seed. The figure omits the neurons and synaptic connections that are not involved in this example. The tonic chord is defined by the notation N_0 , which includes the pitch set $X_0 = \{67, 62, 58, 43\}$, and duration set $Y_0 = \{1, 1, 1, 1\}$, sending to the corresponding parts of the pitch and duration subnetworks.

- Step1: Upon the input of the initial seed notes, the corresponding neurons in the G-minor group of the key cluster are ready to receive the stimuli and emit the spikes (red circle). The pitch and duration values of the seed notes trigger the activation of specific neurons in the sequential memory system (blue and orange cir-

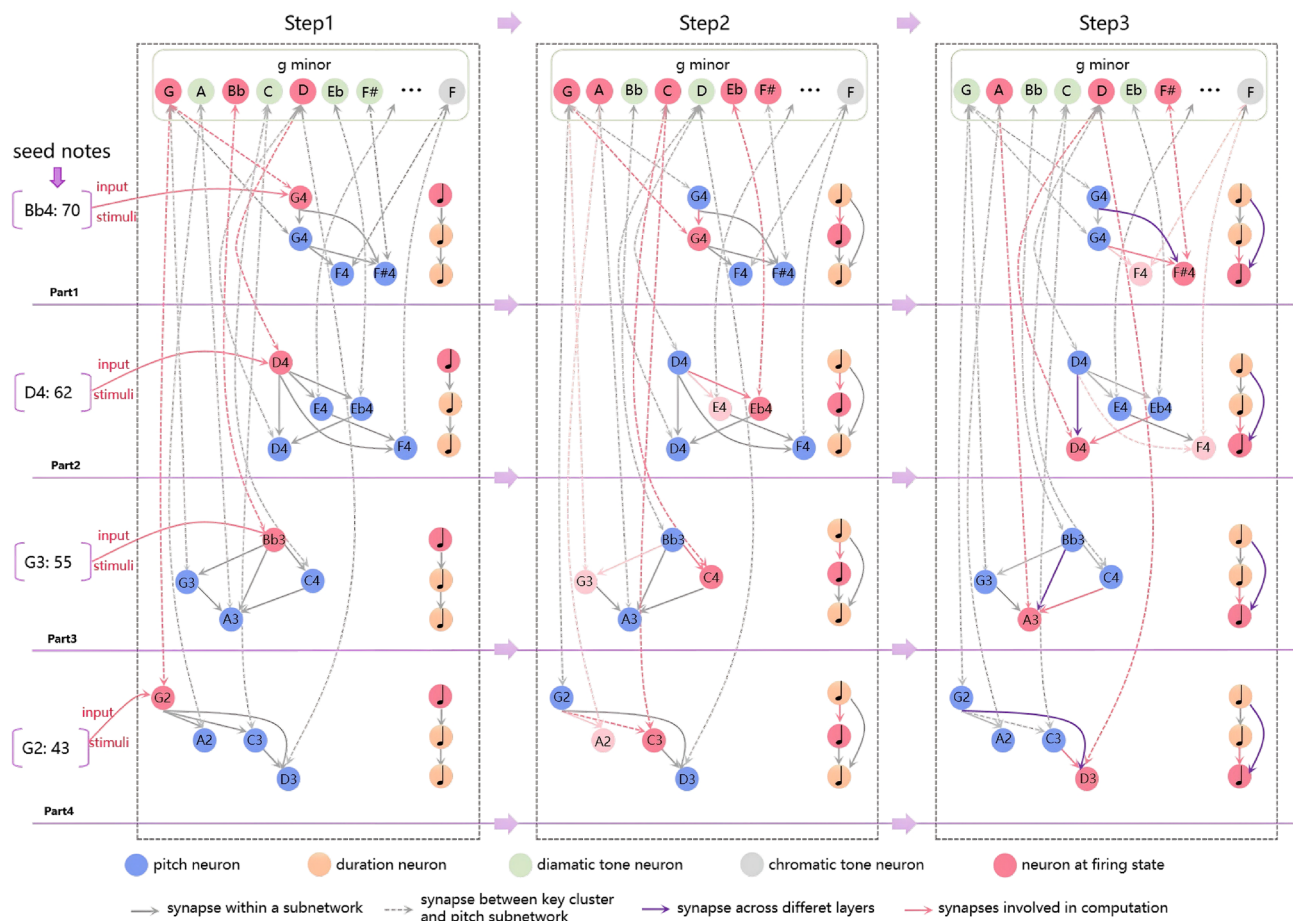


Fig. 4. The generating processes of a musical piece in G minor begin with a tonic chord as the seed. The model receives this seed input and activates the neurons representing pitches that are distributed in four parts, guided by the active neurons that represent G minor, step by step. Neurons and synaptic connections not involved in the generation are omitted for clarity. Green circles represent neurons within the G minor group of the key cluster, blue and orange circles denote the pitch and duration neurons in the sequential memory subsystem, and red circles mark neurons activated at different time steps.

cles), which then propagate through the network to generate the next set of notes. At this step, the neurons are updated by Eqs. (2)–(5). Neurons representing G2, Bb3, D4, and G4 in parts one to four fire (marked by red circles) and propagate their spikes to other neurons.

- Step2: Then, the subsequent neurons of the pitch and duration subnetworks receive and integrate inputs through trained synaptic connections by Eqs. (11) and (12), respectively.

$$I_{ij}^P(t+1) = \sum_s w_s^{TS}(t) + \sum_r w_r^P(t) \quad (11)$$

$$I_{ij}^D(t+1) = \sum_r w_{ri}^D(t) \quad (12)$$

Therefore, neurons in the same layer compete with each other, and we employ the Winner-Takes-All principle to select the most strongly activated neuron as the generated results. As illustrated in Fig. 4, neurons in part two representing E4 and Eb4 are both activated as candidates at this step. However, due to the connection architecture of our trained model between the key cluster and pitch subnetworks, which is similar to KS model, and because the tone E is a chromatic tone in G minor, the synaptic weight between the neuron representing E4 and the neuron representing tone E in G minor group is significantly lower (light pink arrow) compared to that between neuron Eb4 and the corresponding neuron Eb in the G minor group of the key cluster (red arrow). Consequently, the neuron Eb4 emits more spikes and becomes the winner, representing the next generated note in part two. The generating processes are similar in other parts; synapses with high weights are marked by red arrows. C4 and C3 are the final winners in parts three and four. In summary, the neuron with the highest firing rate as the winner in each layer of each part is described by Eq. (13).

$$z = \arg \max_j \left(\sum_t^T S_j(t) \right) \tag{13}$$

Here, z represents the index of the winning neuron, $S_j(t)$ denotes the spike train of neuron j at time t , and T is the total duration over which spikes are counted. Overall, in this step, the generated notes are G4, Eb4, C4, and C3 in the respective parts.

- Step3: The model generates consequent notes by a similar computation process to step two. The difference at this step is that connections across different layers are involved in pitch and duration subnetworks, which are marked by the purple arrows. The final results in this step from part one to part four are F#4, D4, A3, and D3, respectively.

Results and discussion
Datasets

Tonal knowledge can be acquired both implicitly through passive exposure and explicitly through formal instruction. While long-term exposure to music supports the development of intuitive sensitivity to tonal regularities⁴⁹, structured learning, such as through textbooks and targeted exercises, supports the development of abstract, rule-based representations. Neuroimaging studies have shown that explicit musical training enhances cortical encoding of pitch and tonal structure⁵⁰. Inspired by this structured, theory-driven learning pathway, our model is designed to support the symbolic acquisition of modes and keys, in which tonal relationships are explicitly represented, retained, and applied during music generation. To support this framework, we train the model using two datasets:

Sposobin’s harmony textbook exercises (SHTE): To facilitate an in-depth and systematic study of modal and tonal features, this paper introduces a new dataset (<https://github.com/lqnankai/Music-Dataset>) comprising 193 four-part harmony excerpts, carefully selected and annotated from the end-of-chapter exercises in Chapters 4 to 23 of Sposobin’s Harmony Textbook⁵¹. Each excerpt corresponds to a model solution provided by the textbook⁵², originally crafted by music professionals to illustrate specific theoretical principles. The solutions were manually transcribed using the software MuseScore and exported in MusicXML format to support efficient computational processing. Mode and key information were embedded directly into the filenames during transcription to ensure consistent and unambiguous label access. The strength of this dataset lies in its conciseness and theoretical clarity, with each excerpt purposefully designed to highlight a distinct harmonic or modal concept. It includes 96 music pieces in major keys and 97 in minor keys, with the absence of Eb and Ab minor. The number of music pieces in each key of this dataset is detailed in the Table 1

J.S. Bach’s four-part chorales (Bach): To make the model learn more music works, we employ J.S. Bach’s four-part chorales dataset, which is available through the Music21 python package⁵³. It includes 408 famous chorales in *musicxml* format, with 219 in major keys and 189 in minor keys, excluding Db and Gb major, as well as Db, Eb, and Ab minor. A summary of the key distributions is provided in Table 1

Both datasets provide a valuable resource of material for training and evaluating our music generation model, particularly focused on generating four-part harmony in various modes and keys.

Experiments

Alignment With Krumhansl–Schmuckler model

As described in Section 1, the Krumhansl–Schmuckler model (KS model) highlights the significance of pitch classes within each major and minor key. In our model, the synaptic architecture between the mode cluster and the sequential memory system, particularly the pitch subnetwork, mirrors the importance of neural relationships, indicating the critical role of each pitch class within a key. To observe the internal structure of our model, this paper proposes two features, which are listed below:

- Pitch synaptic count (PSC) quantifies the total number of synaptic connections between each neuron representing different pitch classes in the mode clusters and the corresponding neurons in the pitch subnetwork of the sequential memory system. To compute the synaptic count for each pitch class, the 128 minicolumns in the pitch subnetwork, each corresponding to one of the MIDI pitch values (i.e., absolute pitches), are mapped onto the 12 pitch classes (C, C#, ..., B) using the modulo operator. We denote that the k -th neuron represents the k -th pitch class in the mode clusters, the PSC for the k -th pitch class can be computed as Eq. (14).

Keys		C	C#/Db	D	D#/Eb	E	F	F#/Gb	G	G#/Ab	A	A#/Bb	B
Major	SHTE	6	4	11	8	10	13	1	8	8	11	13	3
	Bach	22	–	32	9	10	28	–	59	1	32	25	1
Minor	SHTE	9	6	10	–	11	7	17	17	–	9	2	9
	Bach	10	–	27	–	20	2	7	44	–	49	2	28

Table 1. The number of keys in SHTE and Bach datasets.

$$PSC(k) = \sum_{j=1}^4 \sum_{r \% 12 = k}^{128} \sum_s^N c_{k,jrs} \quad (14)$$

- $PSC(k)$ denotes the total number of synaptic connections between the k -th neuron in the mode cluster, which represents the k -th pitch class in either the major or minor mode, and all neurons corresponding to the same pitch class in the pitch subnetwork, where $k \in [0, 11]$. The variable $c_{k,jrs}$ indicates whether a synapse exists between the k -th neuron in the mode cluster and the r -th neuron in the s -th minicolumn of the j -th part of the pitch subnetwork (see Eq. 7). Here, $j \in [1, 4]$ indexes the four voice parts, $r \in [0, 127]$ represents the MIDI pitch index, and $s \in [0, N)$ denotes the index of neurons within a minicolumn. N is the total number of neurons in each minicolumn.
- Pitch Average Synaptic Weights (PASW) calculates the average weights of synapses between each pitch neuron in the mode/key clusters and the pitch subnetwork. The computation is as Eq. (15).

$$PASW(k) = \frac{\sum_{j=1}^4 \sum_{r \% 12 = k}^{128} \sum_s^N w_{j,r,s}(k)}{PSC(k)} \quad (15)$$

where, $w_{j,r,s}$ is the synaptic weight between the k -th neuron in each neural group of the mode cluster and the r -th neuron in the s -th minicolumn of the j -th part in the pitch subnetwork, $PASW(k)$ denotes the average synaptic weight between the k -th pitch class for each mode and related neurons in the pitch subnetwork, PSC_k is computed as the Eq. (14).

To better compare with the KS model profile, we trained our model on both the proposed SHTE dataset and the Bach dataset. After training, we normalized the scores of Pitch Synaptic Counts (PSC), Pitch Average Synaptic Weights (PASW), and the KS pitch profiles. Figure 5 panels (a) and (b) show the results for major and minor modes based on the SHTE dataset, while panels (c) and (d) show the results from the Bach dataset. In each figure, the purple and green lines represent the average synaptic weights and total synaptic counts across the 12 pitch classes. The pink dashed line shows the reference pitch profile from the KS model. The Panel (e) exhibits the consistent similarities between the structural metrics of our model and the KS psychological model.

For the major mode in both dataset (Fig. 5a,c), the Pitch Synaptic Counts (PSCs) and Pitch Average Synaptic Weights (PASWs) are consistent with the KS model, with the tonic (C) always reaching 1.0 and the dominant (G) scoring nearly as high ($PSC_{SHTE}(G) = 0.98$, $PASW_{SHTE}(G) = 0.95$; $PSC_{Bach}(G) = 0.98$, $PASW_{Bach}(G) = 0.90$). The mediant (E) values also align with the KS profile, while the SHTE dataset shows slightly elevated values for the subdominant (F), reflecting the frequent use of subdominant chords in teaching material. Other pitch classes follow the same general trend as the KS model.

For the minor mode (Fig. 5b,d), the Pitch Synaptic Counts (PSCs) and Pitch Average Synaptic Weights (PASWs) also play the most important role at tonic (C). A key difference, however, is that while the KS model rates the mediant (Eb) higher than the dominant (G), both datasets assign greater weight to the dominant, consistent with its functional role in cadences. Additionally, the SHTE dataset shows an ascending trend for the leading tone (B), attributable to its emphasis on chord-learning exercises.

To quantify the alignment between our proposed metrics and the KS model, we calculate the cosine similarities between each of our proposed metrics (PSC and PASW) and the corresponding KS profiles across the two datasets. Cosine similarity is a widely used metric in mathematics and machine learning for evaluating the similarity between two non-zero vectors. It measures the cosine of the angle between two vectors in an inner product space, capturing their directional alignment irrespective of magnitude. In our context, this metric serves to evaluate how closely the statistical representations of mode features (PSC and PASW) resemble the established pitch-class distributions defined by the KS model. A higher value (1) indicates a stronger alignment with the tonal characteristics captured by the KS profiles, thereby validating the interpretability of our proposed features. As shown in Fig. 5e, the cosine similarities between our model metrics and the KS model are consistently high (>0.9) for both the SHTE and Bach datasets. The results indicate a strong alignment between our model's connection architecture and the Krumhansl-Schmuckler psychological key perception model. The presence of subtle differences indicates that the model captures dataset-specific harmonic characteristics, suggesting that it not only aligns with KS-defined roles for tonic and dominant pitches but also adapts to dataset-specific harmonic nuances, such as the influence of subdominant or leading tones.

We also calculate the cosine similarities between these two metrics of the connection architecture and the KS model for the twelve major and minor keys. The result is provided in Fig. 6.

This figure shows the cosine similarities between two features (PSC and PASW) derived from two datasets (SHTE and Bach) compared with the Krumhansl-Schmuckler (KS) model across 24 musical keys (12 major and 12 minor). The key points of analysis are: The majority of values are above 0.7, indicating a strong alignment between the model's features (PSC and PASW) and the KS model. Specific keys, such as C major, G major, A minor, and C minor, exhibit particularly high cosine similarities across all datasets and features, affirming the model's robustness for these keys; For F# major and F# minor, the cosine similarities are consistently lower (below 0.7), suggesting greater variability in the pitch content of musical pieces in these keys within the datasets. This phenomenon suggests a diversity in pitch usage or less structured tonality in the available pieces for these keys; Zero values are observed for C# major, F# major, C# minor, Ab minor, A# minor, and D# minor. These results indicate that no musical pieces in these keys are present in the corresponding datasets (SHTE and Bach). This absence limits the evaluation of the model's performance for these keys. For the SHTE dataset, the PASW feature tends to have slightly higher similarities compared to PSC for most keys. While for the Bach dataset, both PSC and PASW demonstrate comparable trends, with slightly lower values for some keys compared to SHTE.

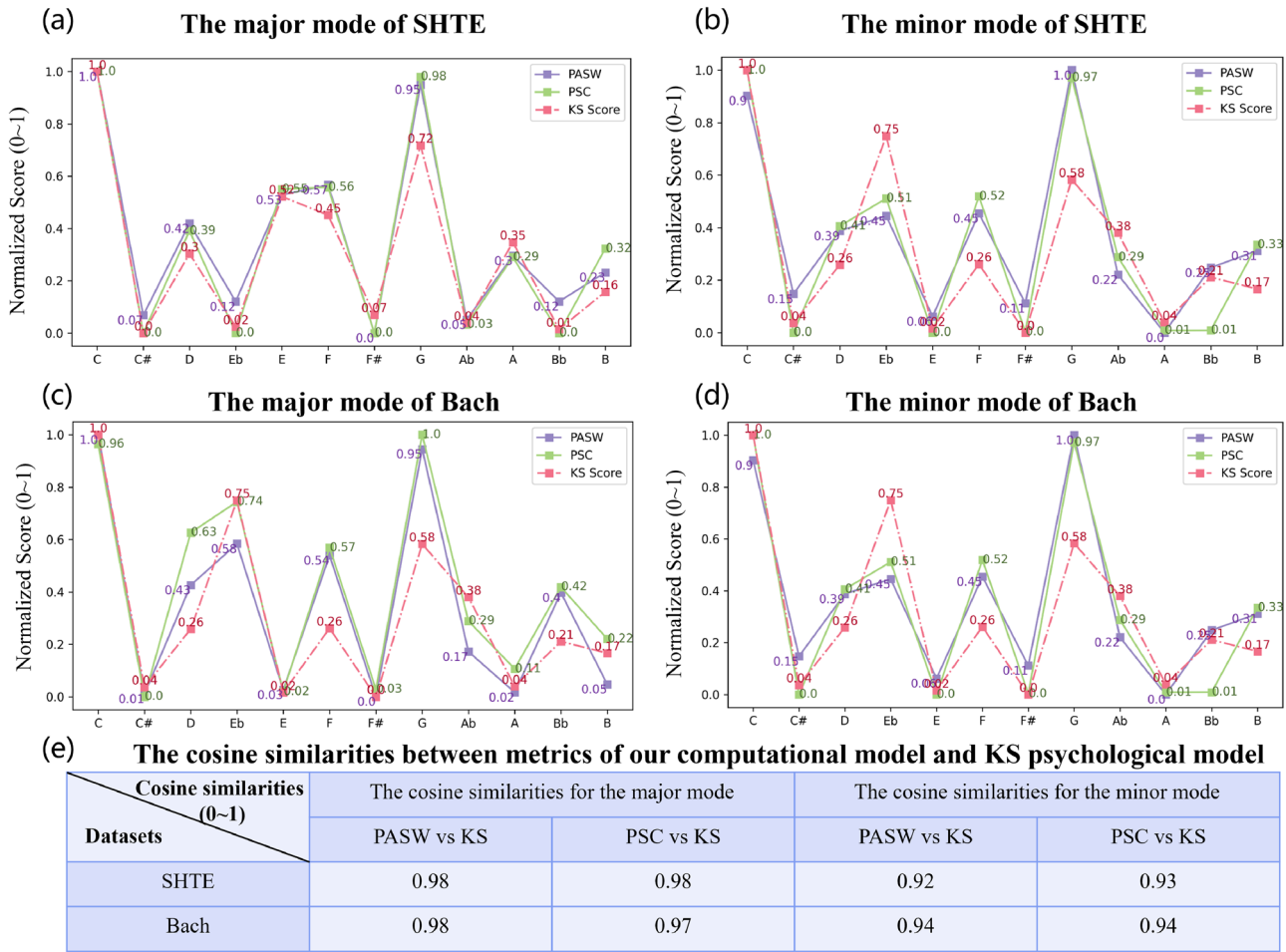


Fig. 5. Comparative analysis of our model and the Krumhansl–Schmuckler (KS) profiles. Panels (a–b) show the average synaptic weight (PASW) and synaptic count (PSC) of neurons in major/minor clusters and pitch subnetworks trained on the SHTE dataset, while panels (c–d) present the same metrics for the Bach corpus. For illustration, pitch-class distributions are aligned to C as the tonic. Panel (e) reports the cosine similarities between these structural metrics and the KS profiles, showing consistently high alignment across datasets.

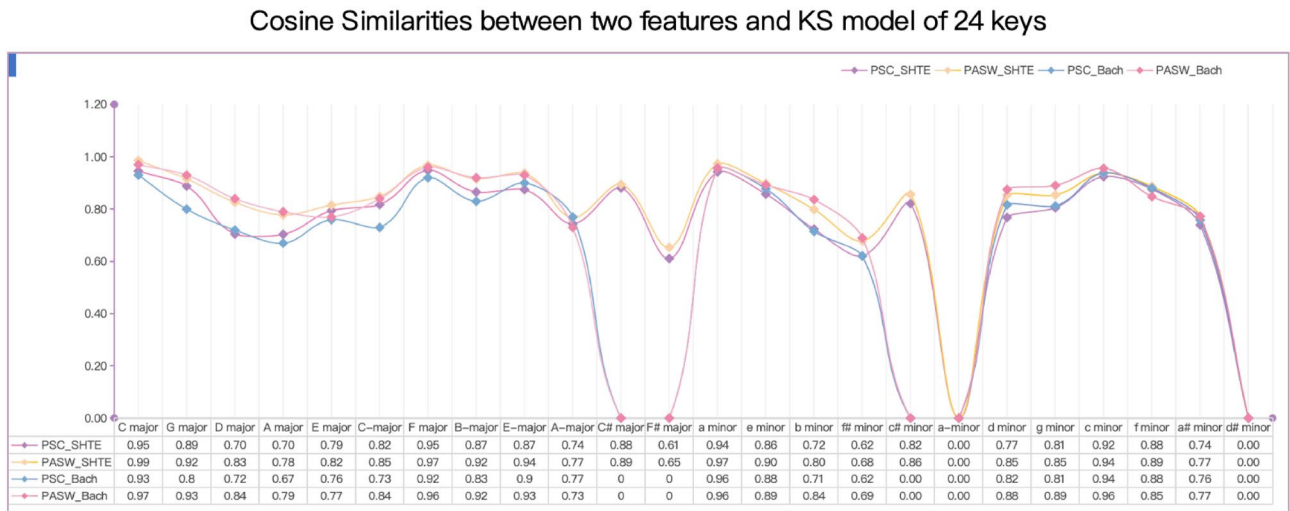


Fig. 6. The results of cosine similarities for 24 keys between the two features (PSC and PASW) and the KS model under the SHTE and Bach Datasets.

Overall, the figure confirms that the developed model's features effectively capture key-related properties similar to human key perception (as modeled by the KS framework). Approximately 91.5% of the values are above 0.7, showcasing the alignment's reliability.

Mode-conditioned generation

In this section, we further assess and analyze the quality of the generated music pieces. We adopt a comprehensive set of objective metrics, including **pitch-related**, **polyphony-related**, and **rhythm-related** features, which are widely used in symbolic music generation evaluation^{54–56}. The pitch-related metrics include: *Pitch Count* (PC)^{54,56}, the number of distinct pitches used in a piece; *Pitch Range* (PR)^{54,56}, the span in semitones between the lowest and highest pitch; *Pitch Interval* (PI)^{54,56}, the average interval size between successive notes; and *Pitch Entropy* (PE)⁵⁵, the Shannon entropy of the normalized pitch histogram. In addition, we introduce the *Diatonic Pitch Rate* (DPR), defined as the proportion of notes belonging to the diatonic scale of the target key, to specifically evaluate tonal consistency, which is a central aspect of our model design. The polyphony-related metrics include *Polyphony* (Pp)⁵⁵, the average number of pitches played simultaneously, and *Polyphony Rate* (PpR)⁵⁵, the ratio of time steps containing multiple concurrent pitches to the total number of time steps. The rhythm-related metrics include *Empty Beat Rate* (EBR)⁵⁵, the ratio of empty beats to the total number of beats, and *Groove Consistency* (GC)⁵⁵, which measures the mean Hamming distance between neighboring measures, reflecting rhythmic regularity.

For baseline comparisons, we adopt several representative ANN architectures, including CNN, RNN, LSTM, GAN, Transformer, and diffusion models, covering the major paradigms in deep generative modeling. The parameters are shown in our supplementary information (see Table S2). From the two original datasets (SHTE and Bach), we randomly selected 10 samples and generated the same number of pieces from each baseline model.

Absolute measurement We first compute absolute measures, including the mean and standard deviation of each metric, to directly characterize the properties of the generated results.

Figure 7a,b report the absolute measure (mean value) of all the models on the SHTE and Bach datasets, respectively. Across both panels, ANN-based models exhibit markedly inflated pitch ranges relative to their respective training data. In contrast, brain-inspired SNNs are close to the data distributions; in particular, our model maintains a realistic pitch range (9.4) and pitch count (7.7; SHTE 9.0, Bach 4.5), indicating stronger alignment with the training material. For the *Diatonic Pitch Rate* (DPR), both training datasets exhibit high DPR values (SHTE 0.96, Bach 0.94), reflecting their strong tonal regularities. Our model achieves a similarly high DPR (0.96), indicating that it effectively captures tonal frameworks, while ANN-based baselines deviate more due to excessive chromatic usage. This finding highlights the model's ability to internalize modal structures in a cognitively meaningful manner. For polyphony- and rhythm-related features, the generated sequences also reach competitive values, further supporting a balance between structural plausibility and variability.

Relative measurement To further assess the similarity between the generated music and the reference datasets, we adopt two complementary metrics: Standardized Euclidean Distance (SED) and Overlap Area (OA). SED quantifies the total distance between two distributions by providing a scale-invariant measure of dissimilarity; lower SED values indicate greater similarity. OA, on the other hand, provides a normalized measure of distributional similarity by calculating the shared area under the probability density functions for one feature, with values ranging from 0 (no overlap) to 1 (perfect overlap). The SED is defined in Eq. (16).

$$d(\mu^{gen}, \mu^{ref}) = \sqrt{\sum_{i=1}^n \frac{(\mu_i^{gen} - \mu_i^{ref})^2}{(\sigma_i + \epsilon)^2}} \quad (16)$$

where μ_i^{gen} and μ_i^{ref} denote the metric means (e.g., PC, PR, and others) of the generated and reference datasets, respectively, σ_i is the standard deviation of the i -th feature in the reference dataset, and $\epsilon = 0.001$ is a

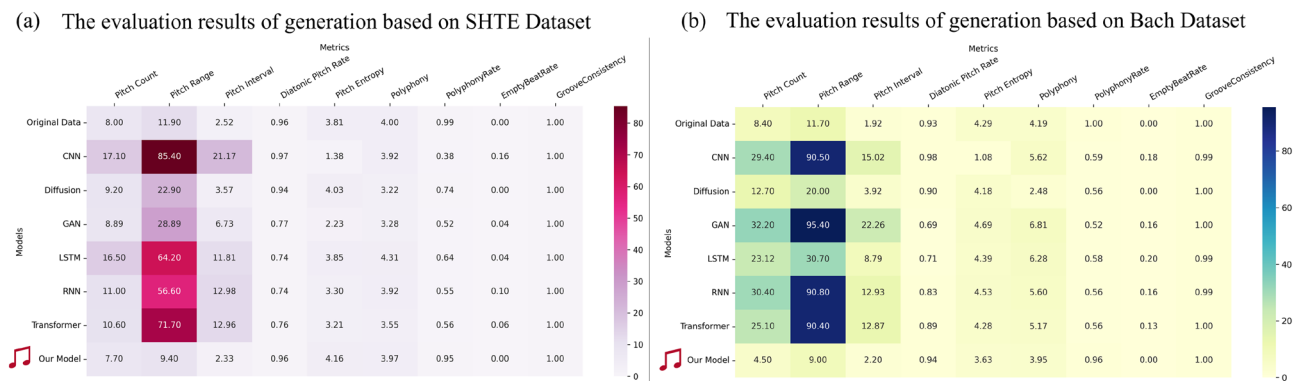


Fig. 7. Comparison of all models on absolute measures: (A) results on the SHTE dataset; (B) results on the Bach dataset.

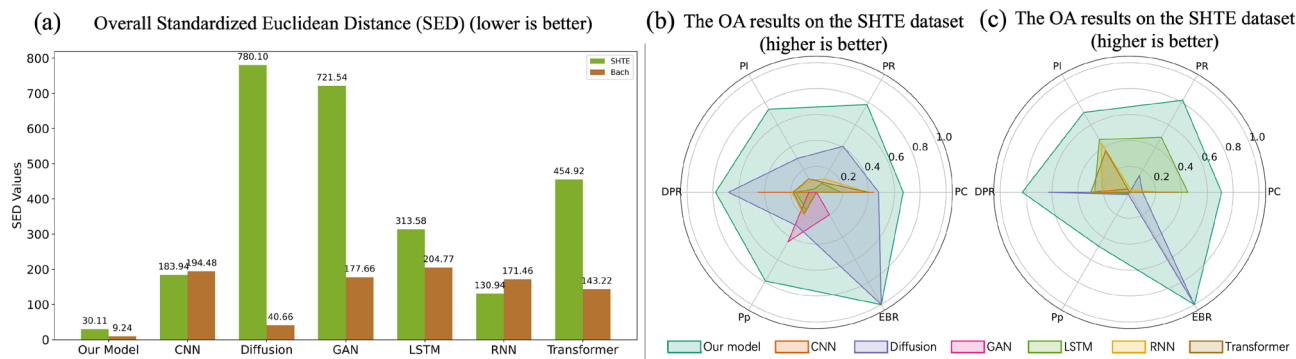


Fig. 8. The results of Standardized Euclidean Distance and Overlap Area across the models, (a) shows the total Standardized Euclidean Distance values across all the models compared with the reference datasets, (b,c) show the key metrics of overlap area values on both datasets.

Dataset	Model	PC	PR	PI	DPR	PE	Pp	PpR	EBR	GC
SHTE	CNN	0.44	0.00	0.00	0.45	0.00	0.00	0.00	0.00	0.09
	Diffusion	0.48	0.41	0.30	0.68	0.74	0.30	0.01	1.00	0.20
	GAN	0.00	0.00	0.00	0.06	0.21	0.44	0.00	0.20	0.71
	LSTM	0.19	0.08	0.03	0.16	0.66	0.16	0.03	0.00	0.06
	RNN	0.40	0.12	0.10	0.19	0.28	0.20	0.02	0.00	0.11
	Transformer	0.40	0.09	0.12	0.18	0.18	0.19	0.01	0.00	0.03
	Our Model	0.67	0.78	0.74	0.78	0.53	0.79	0.10	1.00	0.04
Bach	CNN	0.37	0.00	0.03	0.30	0.00	0.00	0.00	0.00	0.01
	Diffusion	0.10	0.15	0.00	0.63	0.67	0.02	0.01	0.99	0.00
	GAN	0.24	0.00	0.00	0.03	0.23	0.00	0.01	0.00	0.00
	LSTM	0.45	0.49	0.47	0.27	0.57	0.00	0.00	0.00	0.00
	RNN	0.22	0.02	0.44	0.21	0.39	0.01	0.00	0.00	0.00
	Transformer	0.37	0.00	0.37	0.30	0.64	0.01	0.00	0.00	0.00
	Our Model	0.71	0.82	0.71	0.83	0.90	0.48	0.00	1.00	0.20

Table 2. Overlap Area (OA) results on SHTE and Bach datasets (Larger is better).

small constant added to avoid division by zero when $\sigma_i = 0$. Since all feature values are rounded to two decimal places, introducing this parameter ensures numerical stability without affecting the overall scale of the distance. To further evaluate the similarity of each metric, the Overlap Area (OA) offers a normalized measure of distributional agreement by quantifying the shared probability mass between the generated and reference datasets. Its formulation is given in Eq. 17.

$$OA_i = \int \min\{p_i^{gen}(x), p_i^{ref}(x)\} dx \tag{17}$$

where $p_i^{gen}(x)$ and $p_i^{ref}(x)$ denote the probability density functions of the i -th feature (e.g., pitch count, pitch range, etc) for the generated and reference datasets, respectively. The value of OA ranges from 0 (no overlap) to 1 (perfect overlap), with higher values indicating greater similarity between the two distributions. Figure 8a reports the SED results for all models on the SHTE and Bach datasets. Our model achieves the smallest SED values, 30.11 on SHTE and 9.24 on Bach, clearly outperforming all other ANN baselines. On SHTE, the next closest model is RNN (130.94), while Diffusion reaches as high as 780.10, showing a large discrepancy. On Bach, Diffusion performs relatively better (40.66) but still remains far from our model, whereas LSTM yields the largest distance (204.77). These results confirm that our model consistently aligns more closely with the reference distributions, capturing the statistical properties of the training datasets more faithfully than competing architectures. Table 2 summarizes the overall OA values of all metrics across different models on both datasets. While this comprehensive table provides a global view of distributional similarity, we highlight several representative metrics (e.g., Diatonic Pitch Rate, Pitch Count, Pitch Range, Pitch Interval, Pitch Entropy, Polyphony, and Empty Beat Rate) in Fig. 8b,c, which are more closely related to pitch-level learning with modes and keys. On the SHTE dataset, our model achieves the highest OA values on key metrics such as Pitch Count (0.67), Pitch Range (0.78), and Pitch Interval (0.74), significantly outperforming CNN (0.44/0.00/0.00) and RNN (0.40/0.12/0.10). A similar trend is observed on the Bach dataset, where our model again shows superior alignment on Pitch Range

(0.82) and Diatonic Pitch Rate (0.83), while baselines such as CNN and GAN remain close to zero on most pitch-related metrics. These results indicate that our model consistently learns pitch- and mode-related structures more effectively than baseline models, leading to better preservation of musical properties.

Overall, by combining the comprehensive SED evaluation with the OA analysis, the findings indicate that the model effectively captures tonal characteristics and melodic adaptability. This ability to blend established principles with novel expressions underscores its potential for generating diverse, tonally coherent compositions across styles and datasets.

Conclusion and limitation

This work introduces a brain-inspired spiking neural network that integrates biologically grounded mechanisms (e.g., Izhikevich neurons, STDP, minicolumn structures) with symbolic music theory, enabling the model to learn and generate four-part music conditioned on modes and keys. A hierarchical tonal subsystem, functionally inspired by the role of the medial prefrontal cortex in representing modes and keys as schema-like knowledge, is introduced to encode tonal frameworks and modulate learning dynamics and circuit evolution. Notably, the resulting synaptic structures exhibit strong alignment with the Krumhansl-Schmuckler key profiles, indicating that the model internalizes tonal hierarchies in a psychologically plausible manner. Together, compared with current deep learning models, this work demonstrates that brain-inspired spiking neural networks can achieve an initial degree of biological motivation and computational interpretability for symbolic music learning and generation, providing a modeling perspective and a computational foundation for further interdisciplinary research at the level of learning mechanisms.

While the proposed system makes several novel contributions, we also recognize its current limitations. First, this study focuses on pitch-based modal learning and does not incorporate other musical dimensions such as harmonic progression, voice leading, meter, timbre, or dynamics. These components are integral to a more complete characterization of tonal music, and their omission constrains the expressive scope of the present framework. Second, our model emphasizes a structured, theory-driven learning pathway and does not explicitly model implicit tonal learning arising from passive exposure or statistical regularities in musical input. As a result, automatic forms of tonal acquisition typically associated with enculturation and implicit learning paradigms are beyond the scope of this study. Third, we adopt the Krumhansl-Schmuckler model as a reference for evaluating pitch salience and tonal learning. While the KS profile remains a widely used benchmark supported by experimental data, it relies on fixed tonal distributions derived from Western tonal music, which may limit its ability to capture experience-dependent learning and tonal organization across different musical cultures^{57–59}. Accordingly, it is used here as a conceptually grounded baseline for initial validation rather than as a comprehensive model of tonal cognition. Finally, the present study addresses cognitive mechanisms underlying tonal learning and pitch-based structure, whereas music as a human phenomenon also involves emotional, cultural, social, and autobiographical dimensions that lie beyond the scope of the current framework.

In future work, we may extend the proposed framework by incorporating additional neural mechanisms relevant to tonal perception, such as predictive coding and expectation formation, which have been discussed as expectation-based mechanisms relating tonal perception to affective processing^{34,60,61}. We may also extend beyond pitch-based representations to include harmonic structure, enabling higher-level investigations of interactions between harmony, tonality, and affect.

Data Availability

The datasets generated or analysed during the current study are available in the Github repository, https://github.com/BrainCog-X/Brain-Cog/tree/main/examples/Knowledge_Representation_and_Reasoning/musicMemory/task.

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Author contributions

Q.L. designed the study, implemented the code, conducted the experiments, and drafted the manuscript. Y.Z. supervised the project and contributed to the manuscript writing. MHR.T. assisted with code maintenance.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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